

South Atlantic LNG – Role Operating Manual

Position: Mechanical & Process Interface Engineer (Owner's Team)

1. Role Definition

Title: Mechanical & Process Interface Engineer

Reports to: Managing Director

Works with:

- Project Manager (Civil – Construction)
- PMC – ILF Engineers Nigeria
- EPC Contractor – PCC

Core Role

Provide internal technical insight and interface oversight on mechanical and LNG process systems, without replacing PMC authority

- NOT a Project Manager
- NOT a design authority
- NOT approving engineering decisions

2. Role Purpose

The Mechanical & Process Interface Engineer ensures that:

The Owner (SA LNG) maintains technical awareness and visibility across mechanical and LNG systems, while supporting coordination between design and construction.

3. Core Responsibilities

3.1 Technical Awareness & Review

- Review PMC and EPC deliverables at a high level, including:
- Equipment layouts

- Datasheets
- P&IDs (awareness level)
- Identify inconsistencies or potential risks
- Support the MD with simplified technical insights

3.2 Mechanical & Process Interface Monitoring

Focus on critical interfaces between:

- Civil works ↔ mechanical equipment
- Foundations ↔ equipment loads
- Piping ↔ structures
- LNG systems ↔ utilities

Ensure construction readiness aligns with mechanical requirements

3.3 Support to Project Manager (Civil)

Provide technical input to support:

- Installation sequencing
- Equipment delivery readiness
- Site preparation for mechanical systems

PM drives execution — the Mechanical Engineer supports technically

3.4 Participation in Technical Discussions

- Attend PMC/EPC technical meetings
- Ask the right questions
- Ensure key issues are visible to the Owner

3.5 Risk Identification (Technical)

Flag risks related to:

- Cryogenic equipment
- LNG process integration

- Equipment delays
- Mechanical installation challenges

Escalate to PMC — do not resolve independently

4. Role Boundaries (Critical Controls)

The Mechanical Engineer must NOT:

- Approve designs or engineering deliverables (PMC only)
- Issue instructions to EPC contractor
- Override PMC decisions
- Act as Project Manager
- Take ownership of construction execution

This role is advisory and supportive—not authoritative

5. Technical Authority Structure (Very Important)

Area	Owner
Engineering design	PMC
Mechanical systems approval	PMC
Process/LNG systems	PMC
Construction execution	PM (Civil)
Technical awareness	Mechanical Engineer

6. Functional Interfaces

6.1 To Managing Director (MD)

Provides:

- Simplified technical explanations

- Early warning on risks
- Independent technical perspective
- Acts as the MD's technical lens

6.2 To Project Manager (Civil)

Supports:

- Mechanical readiness for construction
- Interface alignment

PM executes → Mechanical Engineer informs

6.3 To PMC (ILF Engineers Nigeria)

- Primary technical interface

Receives:

- Questions
- Observations
- Clarifications

PMC remains final authority

6.4 To EPC Contractor (PCC)

- No direct technical instruction

Interaction only via:

- PM (for coordination)
- PMC (for technical matters)

7. Key Focus Areas (LNG-Specific)

The Engineer should pay close attention to:

- LNG process systems integration

- Cryogenic storage and handling systems
- Compressor and liquefaction units
- Utility systems (power, cooling, gas pre-treatment)
- Mechanical installation sequencing

8. Tools & Tracking

a. Technical Observation Log

- Issue
- Area
- Risk level
- Escalation status

b. Interface Tracker

- Civil ↔ Mechanical
- Mechanical ↔ Process
- Status and readiness

c. Technical Query Log

- Question raised
- To whom (PMC/EPC)
- Response
- Closure status

9. Communication Protocol

With PMC

- Raise technical questions
- Seek clarification
- Validate understanding

With PM

- Provide technical support for execution

- Highlight risks affecting construction

With MD

- Provide:
- Clear, simplified updates
- Key technical risks
- Areas requiring decisions

10. What Makes Him Successful

- Strong technical curiosity
- Knows his boundaries
- Asks the right questions (not too many, not too few)
- Communicates clearly and simply
- Focuses on interfaces and risks, not control

11. One-Line Definition

“Provides technical visibility and interface oversight for the Owner, without replacing the PMC’s engineering authority.”

Final Note

This role is strategically powerful if used correctly:

- Prevents technical blind spots
- Supports the PM without undermining him
- Keeps PMC accountable
- Gives you independent technical confidence